

DECODED EXPLAINED

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India's first mainstream PHEV, and the gap it plugs in the EV ecosystem

Anil Sasi

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THE TOMAHAWK Land Attack The Tomahawk Land Attack Missile is an iconic American long-range, jet-powered, subsonic cruise missile that is used by the United States Navy in ship and submarine-based land-attack operations.

The long-range analogy is perhaps what JSW MG Motor India wanted to play on when it decided to christen its newly-launched Hector Tomahawk plug-in hybrid vehicle (PHEV), alongside an all-electric version of this new SUV, to emphasise range efficiency.

The vehicle is India's first mainstream plug-in hybrid SUV, claimed to deliver over 115km of electric-only range and a combined fuel efficiency

figure of over 40kmpl. MG claims the Tomahawk PHEV can cover over 1,100km with a full tank of fuel and a fully charged battery.

So, what exactly is a PHEV?

To understand this, it's important to examine the textbook definition of an EV — a vehicle that can be powered by an electric motor, which draws electricity from a battery and can be charged from an external source. Beyond this broad classification, though, across most parts of the world, a range of vehicle categories are subsumed as EVs.

Globally, the EV segment covers, besides battery electric vehicles or BEVs, an alphabet soup of some other technologies that incorporate electrifi-

cation in some form: HEVs or hybrid electric vehicles, PHEVs or plug-in hybrid electric vehicles, and FCEVs or hydrogen fuel cell electric vehicles.

All these technologies are part of a broad basket of low-emission vehicular platforms that the Chinese now term as New Energy Vehicles.

Unlike battery electric vehicles or BEVs, which are typically battery electric vehicles such as the Tesla Model S or the Tata Nexon electric that need to be plugged in before the battery is discharged, technologies such as HEVs and PHEVs bring together the best of both worlds. For instance, HEVs, or conventional hybrid electric vehicles, such as strong hybrid variants of the Toyota Hycross Hybrid and the Honda City e:HEV in India, or the Toyota Prius HEV sold

• EV FAMILY: BEVS, HEVS & PHEVS

CONVENTIONAL HYBRID electric vehicles (HEVs) combine a conventional ICE system with an electric propulsion system to produce a hybrid drivetrain that substantially lowers fossil fuel usage.

THE BIG DIFFERENCE is that a PHEV can be plugged into a power source, which offers an external option to charge the battery that is not available on an HEV.

THE PHEVs, or plug-in hybrid electric vehicles also have a hybrid drivetrain that uses a combination of the ICE and electric power for motive power, but is backed by rechargeable batteries, quite like the HEVs.



abroad, combine a conventional internal combustion engine (ICE) system, like in a regular petrol car, with an electric propulsion system to produce a hybrid drivetrain that substan-

tially lowers fossil fuel usage.

The on-board battery in a conventional hybrid vehicle gets charged when the internal combustion engine is powering the drivetrain to produce for-

ward or reverse motion, and this charge from the battery kicks in to offer motive power when it is fully juiced up. There is no need or provision to plug in these HEVs to recharge the batteries.

The PHEVs, or plug-in hybrid electric vehicles, such as the new Tomahawk or the Chevrolet Volt in the US market, also have a hybrid drivetrain that uses a combination of the internal combustion engine (ICE) and electric power for motive power, but is backed by rechargeable batteries, quite like the HEVs.

The big difference therefore is that a PHEV can be plugged into a power source, which offers an external option to charge the battery that is not available on an HEV. A plug-in hybrid car has a comparatively bigger battery and offers a

longer electric range than a traditional HEV. The battery, which gets charged while driving and with regenerative braking, as in an HEV, can be recharged by plugging into a socket if that option is available.

Effectively, a PHEV combines the positive attributes of internal combustion (range on long journeys) with those of plug-in electrics (a cleaner engine that typically costs less to run) in a single vehicle, resulting in a combination of features superior to those of a regular HEV. A bigger battery means that a plug-in hybrid car can be driven completely on 100% electric power far more frequently if it is recharged adequately at a charging point. And unlike a regular hybrid, it does not have to depend entirely on driving or braking for the on-board battery

to get charged, given that external charging is an option in the PHEV that is not there in an HEV. On the flip side, plug-in hybrids are comparatively more expensive to buy than their internal combustion counterparts, and they are also pricier than normal HEVs, given that the on-board battery in a PHEV is bigger than in an HEV.

Running costs for a PHEV, on the other hand, are generally lower, especially if the batteries are charged from an external socket regularly. The important thing is that in the transition to emission-free vehicles, PHEVs offer a genuine progression from hybrid to BEVs, given that they address several concerns with battery electrics — range anxiety being the most important.

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MG Hector Tomahawk PHEV
Battery Electric Vehicles BEV
Hybrid Electric vehicles HEV
Plug in Hybrid Electric Vehicles PHEV
Hydrogen Fuel Cell Electric Vehicles FCEV
New energy Vehicles

A homegrown innovation ecosystem is taking root

Technologies that once arrived in India through imports are increasingly being invented at home through local research institutions, industry, and startups. Three institutional foundations of an innovation economy are now advancing together: public research, corporate research and development (R&D), and deep-tech entrepreneurship. The transformation is yet incomplete, but the architecture is finally visible. Gallium Nitride (GaN) semiconductor technologies, critical for advanced radar, space systems, and next-generation communications, are increasingly being developed domestically, helping close a strategic gap in a tightly regulated, export-controlled field. At the same time, affordable immunotherapies developed in India are expanding access to advanced cancer care and demonstrating the country's growing capacity for biomedical innovation.

Patent filings are often cited as a ready indicator of innovation. The filing numbers continue to show impressive growth. From just over 1,10,000 patent filings in 2024-25, the number grew to more than 1,43,000 in 2025-26, a 30.2% increase. Moreover, domestic applicants now account for almost seven in 10 filings.

However, the sheer volume of applications does not tell the complete story. Even a modest enquiry reveals an innovation story that begins to fray at the edges. The number of patents in force in India (just over 2,40,000 in 2025) is a fraction of those in China (5.7 million), the United States (3.5 million), and Japan (2.1 million); all these are 2024 data.

This suggests low grant rates, high abandonment rates, or a combination of both. Add to this meagre commercialisation, low R&D spending – India still spends just under 1% of GDP on research and development, against about 2.4% in China and 3.5% in the U.S. – and we see how far we are from a truly technology-driven, innovation-led economy.

The emerging pillars

However, there lies a resilient core, and shaping it are three strong and vital strands that promise to anchor India's future. Public institutions with government support have made vital breakthroughs in long-gestation innovation. Private R&D spending exceeded that of all levels of government combined for the first time in 2024. This is expected to climb and account for 55% through 2025-26. Domestic industrial houses are beginning to reap rewards from focused and committed R&D. A maturing startup ecosystem is finding its feet beyond low-hanging service platforms, actively showcasing deep tech and complex manufacturing. That all three strands



Mukundan Chakrapani

Holds a PhD in science and a law degree and is an academic



Gaurav Jain

Founder of Hastin Research and an intellectual property rights expert helping enterprises, startups, and universities transform ideas into intellectual property

Homegrown research and deep-tech enterprise are helping India move beyond its identity as a services provider

are coming of age together bodes well for India.

In March 2023, Defence Research and Development Organisation (DRDO) scientists at the Solid State Physics Laboratory (SSPL) in Delhi and the Gallium Arsenide Enabling Technology Centre (GAETEC) in Hyderabad announced a breakthrough in making gallium nitride (GaN) monolithic microwave integrated circuits (MMICs). The technical know-how to make these circuits, which are critical for military and defence applications, was, by widely reported accounts, refused to India under the offset provisions of the Rafale fighter jet purchase from France. The DRDO is now actively transferring the technology for GaN High Electron Mobility Transistors (HEMTs)-based MMICs to be used in 5G/6G wireless infrastructure, electric vehicles on-board chargers, and renewable energy inverter systems. India is now one of seven countries to have mastered this technology, alongside China, France, Germany, Russia, South Korea and the U.S. AGNIT Semiconductors, a spin-off from the Centre for Nano Science and Engineering (CENSE) at the Indian Institute of Science, Bengaluru is translating homegrown GaN technology to real-world commercial applications.

From implementer to innovator

While long-gestation R&D at public institutions is starting to pay off, there is growing optimism from academia and domestic industry-led R&D as well. The Bharat 6G Alliance (B6GA) is making rapid progress in its stated aim of contributing 10% of the global 6G patents by 2030. According to a review meeting by Union Minister for Communications Jyotiraditya Scindia, B6GA members have made more than 7,700 patent filings across 5G and 6G technologies, including over 4,400 foreign filings. But, these are applications, not yet grants or declared standard-essential patents, and their true impact remains to be seen. However, taken together with almost 3,000 technical contributions to 3GPP last year, a 15-fold increase from 2020, these figures point in the right direction. The World Intellectual Property Organization's 2025 Patent Cooperation Treaty rankings showed that Jio Platforms Limited had broken into the top 20 international patent filers for the first time, rising more than 300 places to be placed 19th overall.

While patent numbers are only a part of the story, participation in standards bodies and intellectual property ownership increasingly determine who captures value from secure communication networks. India is securing ownership of critical digital infrastructure and is moving from being a standard-implementer to a standard-setting innovator.

There is also a marked shift in India's startup ecosystem. Buoyed by the government's commitment to Research, Development, and Innovation (RDI) and private capital matched by initiatives such as the more than \$2.5 billion in deep-tech commitments made by members of the India Deep Tech Alliance (IDTA), Indian startups are aiming farther and higher than before. Pixxel Space, founded by two Birla Institute of Technology and Science (BITS) Pilani alumni, is carving a niche in hyperspectral imaging technology with six satellites in orbit and a full constellation of 18 to 24 planned. The Vikram-1 low earth orbit launch by Skyroot Aerospace and the IIT Madras nurtured Agnikul Cosmos's parallel push towards fully reusable launch vehicles promise to strengthen India's role as a frugal space technology leader.

Startups such as ImmunoACT, incubated at IIT Bombay in partnership with the Tata Memorial Centre, are expanding access to advanced and affordable health care. NexCAR19, India's first indigenous CAR-T cell therapy, has delivered effective cancer treatment at a tenth of typical treatment costs. Leveraging the Biotechnology Industry Research Assistance Council (BIRAC) early-stage support, Bengaluru-based Remidio Innovative Solutions is at the forefront in the fight against preventable blindness, such as diabetic retinopathy and glaucoma, through smartphone enabled retinal imaging systems and AI.

The direction is clear

These examples do not by themselves make India an innovation superpower but demonstrate that the three strands are coming together to weave an enduring innovation ecosystem. The GaN story illustrates the emerging innovation pipeline. Government laboratories developed the foundational capability, academic institutions nurtured the talent and research ecosystem, and startups such as AGNIT are now carrying the technology into commercial markets.

However, sustaining these three strands will demand more work, from standardising tech-transfer terms for publicly funded IP and expanding examiner capacity at the patent office, to funding the translational gap between a granted patent and a first paying customer. Yet, as we look through the innovation lens, the direction is clear. India is gradually shedding its identity as a back-office service provider and strengthening the institutions that enable nations to generate their own technology, intellectual property, and industrial capabilities. The journey remains unfinished, but the contours of an innovation economy are beginning to form.

R&D spending (% of GDP)

China 2.4%

US 3.5%

India 0.7%

Gallium Nitride (GaN)
Critical for radar space systems

Patent filing
Intellectual property rights
India's startup ecosystem

1. Public research institutions
2. Corporate R&D
3. Deep-tech startups

The Himalayan Monal- state bird of Uttarakhand
Calls are important for communication, reproduction
and social behaviour.

Tourism and traffic; increasing human made noise

A Himalayan bird's calls are changing due to human noise: Why this matters

Vidheesha Kuntamalla
Dehradun, September 18

THE HIMALAYAN monal, the state bird of Uttarakhand, is a striking-looking landfowl found across the Himalayan mountains, including in Jammu and Kashmir, Himachal Pradesh, Uttarakhand and Sikkim.

While the ground-dwelling bird is not endangered, a study carried out by researchers at Hemvati Bahuguna Garhwal University (HBGU) has found that the bird is changing its calls — because tourism and traffic are making its surroundings noisier. This could have implications for its success in mating, though the researchers, Rajeev Lochan, Dinesh Kumar Sharma and Anand Kumar, have not established this.

What did the study show?

To study the monal's calls, the researchers used passive acoustic monitoring devices, along with a microphone, across different seasons. They compared two areas within the Kedar-nath Musk Deer Sanctuary — the Madmaheshwar Valley, which has relatively little human activity, and Chopta Valley, which



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A Himalayan monal.

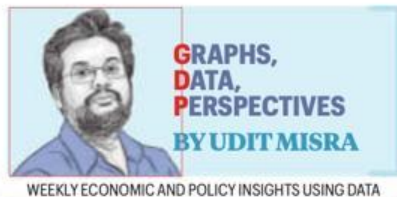
Chopta has a tendency to call with more frequency and amplitude. In Madmaheshwar valley, it was relatively less. The reason is that there is human-made noise, either through transportation or the heavy footfall from the pilgrimage," co-author Anand Kumar, an assistant professor at HBGU's Department of Zoology, told *The Indian Express*.

The study found that this difference was particularly strong at the higher end of the birds' call frequencies. The researchers say this suggests that the monal has some ability to adjust its calls to its surroundings.

Why is this important?

For the monal, calls are important for communication, including reproduction and social behaviour. The researchers worry that increasing human noise could make such communication more difficult. "The male makes the mating call and the female responds to that call," co-author Lochan told *The Indian Express*. But with more tourism and noise, "it will not be able to reach females, because it will be masked with noise." The researchers say more work is needed to understand what the changes in calls mean for the monal's breeding and survival. The study does not say that only tourism is affecting the monal's calls, but calls for better management of noise in heavily visited areas. The answer, they say, is not to stop tourism, but to manage how human activity affects these landscapes.

Hawkish or dovish? The varying interpretations of US Fed rate hike



THE FEDERAL Reserve, which controls US monetary policy, has decided to raise the Federal Funds Rate (FFR) by 25 basis points (one basis point is one-hundredth of a percentage point). It is now in the 3.75% to 4% range, up from 3.5% to 3.75%.

Raising the FFR makes it costlier for banks to borrow from each other and, thus, makes all loans extended by US banks costlier. Higher cost of credit drags down economic activity as it discourages overall demand and helps cool down prices in the economy. This decision is typically taken when a central bank is concerned about curbing inflation. It is keenly watched worldwide because tweaking credit costs also affects stock markets.

The US Fed has a dual mandate: Pursuing both price stability and maximum employment. For price stability, the Fed has a target of 2% inflation. Employment has no numerical target. Often, the Fed's

focus is either more on curbing inflation or boosting employment, depending on what it thinks is the bigger concern. What stood out about the Fed's decision was that it could make for a variety of interpretations.

Hawkish

A hawkish central bank or hawkish policy decision is primarily focused on maintaining price stability and not letting inflation rise above 2%. An increase in the FFR — the first in three years — suggests that new Fed Chair Kevin Warsh has seen enough price rises with the West Asia crisis and tariffs. He is even willing to go against US President Donald Trump by raising rates, months before the crucial midterm elections.

Warsh noted that while "economic activity is expanding" and "the unemployment rate has changed little", "inflation had remained elevated".

Not so hawkish

Another interpretation would argue that this policy isn't all that hawkish. The table provides the Fed's Summary of Economic Projections (SEP).

Four times a year, the Federal Reserve releases a summary of Federal Open Market Committee (FOMC) participants' projections for GDP growth, the unemployment

• US Fed's Summary of Economic Projections

Variable	2026	2027	2028	2029	Longer-run trend
Change in real GDP	2.3	2.4	2.2	2.1	2.0
June projection	2.2	2.3	2.2	—	2.0
Unemployment rate	4.1	4.1	4.1	4.1	4.2
June projection	4.3	4.3	4.2	—	4.2
Inflation	3.7	2.3	2.1	2.0	2.0
June projection	3.6	2.3	2.0	—	2.0
Federal Funds Rate	4.1	4.1	3.9	3.6	3.2
June projection	3.8	3.6	3.4	—	3.1

SOURCE: U.S. FEDERAL RESERVE

Two stances

- A hawkish policy decision is focused on maintaining price stability and not letting inflation rise.
- A dovish policy prioritises growth or job creation.

rate, inflation, and the appropriate policy interest rate. The FOMC is the 12-member entity that decides on monetary policy.

The table shows the median FOMC projections. The FOMC expects economic growth to be stronger than its June assessment, even higher than the long-run trend. Unemployment projections, too, are expected to be better than the long-run average.

Inflation is expected to worsen, and the FOMC does not expect it to return to its 2% level before 2029. US inflation was last under 2% in February 2021.

Given this projection, a hawkish central

bank would have signalled many more interest rate hikes, but the Fed has only pencilled in a single 25-basis-point rate hike over the next year. Many in the markets believe the Fed will have to undertake an additional three rate hikes in the next year.

Actually dovish

For these reasons, one could argue that the interest rate hike was not hawkish enough and points to a rather dovish stance. Considering that the Fed is unlikely to have chosen the October policy review for a rate hike, coming too close to the midterms, it follows that it chose the bare minimum now.

Not dovish enough

While some argue about the stance, at least one man vociferously believes that this policy decision was not dovish at all.

Trump criticised the Fed decision on Truth Social: "Interest Rates in the United States should be 1%, or less, because we are the Best Credit in the World — BY FAR. Our Country is BOOMING with new Investment... LOWER THE INTEREST RATES... AND FAST!" Earlier, too, Trump berated Warsh's predecessor, Jerome Powell, for being "too late" to cut interest rates.

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The US Federal Reserve

US Monetary Policy

Dual mandate: pursuing both price stability and maximum employment

Federal Funds Rate (FFR) increased

Loans are costlier, economic activity down
Overall demand down, cool down prices
Curbing inflation, affects stock markets

Hawkish policy decision is focused on maintaining price stability and not letting inflation rise.

Dovish Policy prioritises growth or job creation

Federal open market committee (FOMC)
12 member entity

Fed "Price stability + Maximum employment"

RBI "Price stability is the primary objective, while keeping growth in mind."

UPSC Prelims 2021 Fed + RBI + Government Bond Yields
Indian Government Bond Yields are influenced by which of the following?

- 1.Actions of the United States Federal Reserve
- 2.Actions of the Reserve Bank of India
- 3.Inflation and short-term interest rates

Select the correct answer using the code given below:

- (a) 1 and 2 only
- (b) 2 only
- (c) 3 only
- (d) 1, 2 and 3**

UPSC Prelims 2022 — Directly on US Federal Reserve

Consider the following statements:

1. Tight monetary policy of US Federal Reserve could lead to capital flight.
2. Capital flight may increase the interest cost of firms with existing External Commercial Borrowings (ECBs).
3. Devaluation of domestic currency decreases the currency risk associated with ECBs.

Which of the statements given above are correct?

- (a) 1 and 2 only**
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Deep-sea discovery, environmental responsibility

India is standing at the threshold of a new frontier. But the most important question is not how deep we can go, how much we can mine, or how much wealth lies beneath the ocean floor. It is more fundamental: Should everything that can be technologically exploited necessarily be exploited? The question is particularly important as India strengthens its position in deep-sea exploration and develops the scientific and technological capabilities to access one of the least understood parts of the planet.

A growing deep-sea footprint

India is not new to seabed mineral exploration and holds three International Seabed Authority (ISA) exploration contracts covering approximately 95,000 square kilometres across the Central Indian Ocean Basin, Central Indian Ridge and Carlsberg Ridge in the international seabed area. Its exploration has identified an estimated 366 million tonnes of polymetallic nodules containing nickel, copper, cobalt and manganese, alongside extensive environmental and biodiversity studies.

Under the Deep Ocean Mission, India is developing deep-sea mining technology, underwater robotics and the MATSYA-6000 human submersible, while the National Institute of Ocean Technology has tested a mining machine at depths of about 5,270 metres. At the same time, biodiversity surveys across 19 seamounts have studied around 1,300 deep-sea organisms, including nearly 23 species reported as new to science. These achievements also impose a profound responsibility to decide whether technological ability should ultimately lead to exploitation.

Technological progress can create a dangerous assumption: if we can exploit something, we should. But the deep ocean challenges this logic. It is a complex and poorly understood ecosystem, and India's own biodiversity research continues to reveal previously unknown organisms. We cannot confidently predict the consequences of



P. Ragavan

Marine biologist with over a decade of professional experience in the conservation and management of coastal and marine environments

India has an opportunity to become a global leader not only in deep-sea technology but also in deep-sea ecological governance

large-scale seabed disturbance while our understanding of these ecosystems remains incomplete. This is not an argument against science, but for more science before potentially irreversible intervention. Deep-sea minerals may have legitimate strategic importance for India's renewable-energy, electric-mobility and advanced-manufacturing ambitions. However, strategic importance cannot automatically become ecological permission. Before considering extraction, India should ask whether the minerals are genuinely necessary at the proposed scale, whether alternatives exist, and whether demand can be reduced through recycling, efficiency, substitution and a circular economy. Necessity must be demonstrated, not presumed.

India's investment in deep-ocean science need not imply a commitment to commercial mining. Exploration can advance knowledge, biodiversity assessment, environmental baselines and technology without leading to exploitation. Mining, however, would physically disturb the seabed. Therefore, adaptive management alone cannot guarantee ecological protection. India has an opportunity to adopt a stronger principle before exploitation begins: the question should not be "How can we mine with minimum damage?" but "Do we need to mine at all?" Since India's ISA activities remain at the exploration stage, this is the right time to establish a high ecological threshold for any future decision to exploit.

Science must also teach us to say 'no'

The deep ocean is now emerging as the next frontier for economic growth. Yet, we expect the ocean to regulate climate and support biodiversity while extending industrial activity into its least understood ecosystems. The ocean should not become a place where the environmental costs of human consumption are simply relocated. Scientific progress should not mean making more of nature exploitable and

then calling it sustainable. True sustainable development must reduce dependence on extraction through lower demand, resource efficiency, recycling, substitution and circular consumption – principles reflected in India's Mission LiFE. Science must therefore help us use resources responsibly and, equally importantly, determine when we do not need to use them at all. This calls for a broader understanding of scientific progress. Science should not merely open new frontiers for exploitation; it should also reveal which frontiers are best left untouched. The ability to say "we do not need to go there" may itself represent a higher level of scientific and civilisational maturity than simply proving that we can

India can lead a different model

India has an opportunity to become a global leader in deep-sea ecological governance. India could take the bold step of recognising the deep ocean not merely as a mineral reserve, but as a precious ecological asset whose value may exceed that of its mineral wealth. It could use the knowledge gained through the Deep Ocean Mission to conclude that some poorly understood and potentially irreversible ecosystems are better left undisturbed. Progress is not how much nature we can convert into resources, but how much we can preserve while meeting human needs. This would reaffirm India's long-standing civilisational tradition of living in harmony with nature.

The defining environmental approaches of this century – Nature-based Solutions, the SDGs, the Circular Economy and Mission LiFE – share a common principle: nature is the capital on which development depends. They call for conserving ecosystems, reducing resource consumption and ensuring responsible development. Science and technology must therefore tell us not only how far we can go but also where we must stop. Our future depends on preserving the natural capital that sustains development.

Deep sea exploration

International Seabed authority (ISA)

95000km square, central Indian ocean

Central Indian Ridge & Calsberg Ridge

Polymetallic nodules (cobalt, Nickel, copper)

Deep ocean Mission – Samudrayan

Matsya-6000- Human submersible

India's Mission LiFE, SDGs

Circular economy, Nature based solutions

Living in harmony with nature

Intergenerational equity

Thank you

“The price of success is daily effort”



gallantias@gmail.com



+91 9746035110